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EnergyLens — Conversation & Q&A Prep Pack

Event: metiundo “Energiewende konkret!”, 1 July 2026, Berlin **Purpose:** Answer questions confidently, speak the right English/German terms, and frame current limits honestly as deliberate, capacity-aware choices with a clear roadmap. You are an **energy engineer + founder** — this pack backs that identity.

TR — **Nasıl kullan:** Bölüm 2 (yöntem) + Bölüm 6 (sözlük) = “ne yaptığını bilen” havası.
Bölüm 4 = her şirketle ne konuşacağın. Bölüm 5 = zor sorulara dürüst cevap. Ezber değil, mantığı kafanda otursun yeter.

1. EnergyLens in one minute

EnergyLens is a **data-light, hardware-free, estimation-first energy + EU-compliance intelligence layer for building portfolios** (commercial and residential, Germany first). It starts from data a building already has — bills, existing meter readings, the energy certificate (EPC), age/type/size — and produces energy intensity, carbon, compliance risk (EPBD/CRREM) and ranked retrofit measures with subsidy-adjusted ROI. **Every number is a range with its assumptions**, and is replaced by measured values as real data arrives.

Five core focuses: (1) hardware-free estimation engine, (2) energy efficiency & cost, (3) carbon & ESG (ESRS-E1-aligned), (4) compliance & regulatory risk — the wedge, (5) portfolio CapEx prioritization.

TR — **Not:** Bu odadaki lider kartın = uyumluluk + portföy-kararı + donanımsız tahmin.
Flexibility/trading iddiası YOK; o oyuncuları tamamlıyorsun.

2. How it actually works — methods, assumptions, standards

(This is the backbone. If someone asks “how do you calculate X without hardware?”, answer from here.)

2.1 The estimation engine

- **Inputs:** utility bills / annual consumption, building type, floor area (Wohn-/Nutzfläche), construction year, EPC, plus public building footprints (OSM/official) and public EPC data.
- **Method:** building **archetype profiles** (typical energy use by type + vintage) → a **provisional baseline EUI range** → corrected for **weather (degree-days)** and the actual bills you provide.

- **Output:** a defensible first picture as a **range + stated assumptions**, flagged “Estimated”; replaced by measured values as real data comes in.
- **Say it:** *“It’s a physics- and archetype-based estimate, not a black box. You get a range with the assumptions written next to it, and it tightens as real data arrives.”*

TR — **Not:** “Estimation-first” = sensör beklemeden, eldeki veriyle ilk tabloyu kurmak; her değer aralık + varsayım, ölçüm gelince yerini ölçülen değer alır.

2.2 Energy intensity & benchmarking

- **EUI = Energy Use Intensity = kWh/m²/year** (the core efficiency metric). Compared against archetype/peer bands and EPC classes.
- **Weather normalization:** consumption is normalized with **HDD (Heating Degree Days)** so a cold year doesn’t look like waste. Formula pattern: $\text{normalized} = \text{raw} \times \text{HDD}_{\text{normal}} / \text{HDD}_{\text{period}}$.
- **Standard:** EPC bands (Energieausweis A-H); archetype EUI bands (e.g. office $\approx$ 120-220 kWh/m²/yr — *indicative*).

TR — **Not:** EUI = binanın “yakıt tüketimi/100km”si. HDD ile düzeltme = soğuk yılı israp gibi göstermemek için.

2.3 HVAC & building envelope (Page 7 / gold_hvac_analytics)

- **Heating/cooling/ventilation split:** estimated with an **HDD/CDD proportional model** (ventilation $\approx$ fixed 15%) when no sub-meters exist.
- **Heat loss:** simplified **EN ISO 13370** transmission method: $U_{\text{composite}} \times \text{envelope_area} \times \text{HDD} \times 24 / 1000$.
- **Insulation score (0-100):** composite U-value vs **GEG 2023** reference U-values (100 $\approx$ passive-house standard).
- **Heat pump: COP** (Coefficient of Performance, instantaneous) and **SCOP** (Seasonal COP, Oct-Apr rolling); only meaningful where a heat pump exists.
- **Envelope area:** currently estimated (square-plan, $\sim$ 3-storey assumption) → tightens with real BIM/DWG or a quick survey.
- **Say it:** *“HVAC is an assessment layer — I estimate the heating/cooling split, transmission heat loss via ISO 13370, and an insulation score against GEG reference U-values, then flag renovation priority. I don’t control systems; that’s the operational layer.”*

TR — **Not:** Bizim HVAC = değerlendirme (hangi bina kötü, önce hangisi). Green Fusion = kontrol (sistemi fiilen çalıştırıyor). Bunu karıştırma.

2.4 Carbon & ESG

- **Scopes:** Scope 1 (on-site combustion, refrigerants), Scope 2 (purchased electricity/heat), Scope 3 (upstream).
- **Scope 2 two ways: location-based** (grid average) vs **market-based** (contracts/residual mix). DE grid $\approx$ **363 g CO₂/kWh (2024, UBA)**; DE **residual mix** is much higher ($\sim$ 0.7 kg/kWh) — matters for market-based.
- **Reporting: ESRS-E1-aligned** / voluntary **VSME**. (Always say “ESRS-E1-aligned,” never “CSRD-compliant.”)

TR — **Not:** Location vs market-based ayrımı sorulabilir — location = şebeke ortalaması, market-based = senin sözleşmen/residual mix. İkisini bil.

2.5 Compliance & regulatory risk (*the wedge*)

- **EPBD + national GEG (Gebäudeenergiegesetz):** minimum energy performance standards (MEPS) tightening around **2030 / 2033**.
- **CRREM:** decarbonization pathways → the year a building “**strands**” (falls below the carbon target). (*CRREM pathways/factors are open-access; embedding in a commercial product needs a CRREM License Partner.*)
- **EU Taxonomy:** building eligibility by EPC/efficiency (thresholds under revision).
- **Carbon pricing on buildings: nEHS** today (~55–65 €/t), **EU ETS2** starting ~**2028** → the rising “cost of doing nothing.”
- **GEG §71a:** building-automation requirement (this is what Green Fusion is TÜV-certified for — you screen for it, you don’t certify).
- **Residential: HeizkostenV / HKVO** monthly consumption info (**UVI**) mandatory from **1 Jan 2027** (with a ~3% billing-cut penalty for non-compliance); cost allocation typically **70/30** (consumption/area).

TR — **Not:** “Cost of doing nothing” = uyum yapmamanın yükselen maliyeti (ceza + karbon fiyatı + değer kaybı). Senin satış mantığının kalbi bu.

2.6 Retrofit ROI & subsidies

- **Ranked measures** with payback + a **MACC** (Marginal Abatement Cost Curve) view: cheapest CO₂ savings first.
- **Subsidy-adjusted ROI: BAFA** heat-pump funding (base ~30%, up to ~70% with bonuses) and **KfW/BEG** programs reduce net CapEx.
- **Retrofit physics:** savings from a measure $\approx \Delta U \times \text{area} \times \text{degree-days} \div \text{system efficiency}$ (Energieberater-style).

TR — **Not:** “Subvansiyon-ayarlı ROI” = sübvansiyonu düşünce gerçek geri-dönüş süresi. BAFA/KfW’yi anmak seni ciddi gösterir.

2.7 Forecasting — why rule/physics-based (for now)

- Current forecast/estimate is **deterministic (archetype + degree-days + bills)**, not trained ML — **on purpose:** it’s **transparent, auditable and explainable**, which is exactly what compliance and investment decisions need. A black-box ML number is a liability in a compliance report.
- **ML comes later, data-first:** once real multi-building data exists (from pilots), ML is added **only where it beats the rule-based baseline** (e.g. consumption disaggregation, anomaly detection). Short-term load/flex forecasting (the ML-heavy kind) is The Mobility House’s core, not ours.
- **Say it:** “*It’s deterministic and transparent by design — for compliance you need numbers you can defend. I’ll add ML later on real data, only where it clearly beats the physics baseline.*”

TR — **Not:** “ML kullanıyor musun?” gelirse savunmaya geçme — şeffaflık bilinçli bir seçim, de. Asıl darboğaz model değil, gerçek veri.

3. Current state vs. roadmap (*honest, capacity-aware framing*)

Use this whenever something isn’t fully automated — turn a limit into a deliberate choice + a plan.

Topic	Today (bootstrapped, ~€0)	With a pilot / funding / capacity
Compute	Static snapshots; gold tables mirrored to Postgres; Microsoft Fabric used for batch only	Always-on Fabric capacity (F-SKU); live refresh
Real-time / IoT	Off (€0); architecture ready (“decoupled” — Azure Event Hubs + hot store)	Switch on per paying IoT customer (scale cost, not upfront burn)
Climate/HDD	Placeholder factor (1.0) on some paths	Wire real HDD per postal code (DWD Open Data / Open-Meteo)
Data connectors	Generic bill/CSV/Excel ingest; vendor exports mapped manually	Productized connectors (Techem/ista/Aareon) once it’s a repeated paying pattern
EPC	Customer-provided / manual entry	Automated import as the DE central EPC register matures
Deep BI	Founder-run “bridge” into my Fabric	Customer self-serve, embedded
Data realism	Synthetic/sample data, every output QA’d by hand	Real-building data from pilots → tighter, validated models

- **Say it:** *“Right now I run it lean — Fabric for batch, real-time switched off — because compute capacity is a real cost and I keep burn at zero until a paying customer needs it. The decoupled architecture is already there; I switch on the real-time layer per customer. It’s a deliberate cost choice, not a missing piece.”*

TR — **Not:** Bu tablo senin en güçlü kozun — “param/kapasitem yok” demek yerine “bilinçli olarak €0 tutuyorum, mimari hazır, müşteri gelince açıyorum” diyorsun. Olgun founder havası.

4. Per-company playbook

Each: their angle · likely questions + your answers · what they’ll probe (calc/standard) · now→future · TR note.

4.1 metiundo (host) — Dennis Nasrun, CEO

Their angle: smart-metering-as-a-service; they own the data/hardware layer (electricity/gas/heat/water), €40M (Octopus), hiring. **Likely questions & answers:** - *“How do you work without a smart meter installed?”* → *“Estimation-first: archetype + degree-days + bills + EPC, as ranges. When your meters come online, measured values replace the estimates.”* - *“How would you use our data?”* → *“You own the data pipe; I’m the compliance and decision layer on top — EPBD/CRREM, retrofit prioritization. Your data makes my layer sharper; my layer makes your data more valuable to the owner.”* - *“What’s your data architecture?”* → *“Medallion — bronze/silver/gold on a lakehouse, building_id as the key, per-building and per-unit grain.”*
They’ll probe: data model cleanliness, MsbG / iMSys / Smart Meter Gateway awareness,

market communication. **Now→future:** “Today I ingest exports/CSV and bridge into Fabric; with funding I’d take a structured API feed from a metering partner and switch on the real-time path.”
> TR — **Not:** metiundo = partner + işveren. “Rakip değilim, üst katmanım” mesajı net olsun. CV değil, ilgi göster + LinkedIn.

4.2 Green Fusion — Paul Hock, MD

Their angle: AI heating optimization + control, GreenBox hardware, GEG §71a TÜV-certified, ~16% measured savings, €12M. **Likely questions & answers:** - “Do you control heating?” → “No — I assess and screen. I tell the owner which buildings to prioritize; you optimize and run them. Different layers.” - “How do you estimate HVAC without sub-meters?” → “HDD/CDD proportional split, transmission heat loss via ISO 13370, insulation score against GEG reference U-values — all as ranges, replaced by measured data where it exists (including yours).” - “What about §71a?” → “I flag it at portfolio level; I’m decision-support, not a certified building-automation system. That’s your strength.” **They’ll probe:** whether you claim to control (don’t), COP/SCOP understanding, GEG §71a. **Collaboration line:** “You control and optimize; I sit one layer above with screening and EPBD/CRREM compliance. I hand you a prioritized target list, you execute, I track the compliance impact — complementary, not competing.” **Now→future:** “Envelope/heat-loss is estimated from geometry assumptions today; with a survey or your operational data it becomes measured.” > TR — **Not:** En yakın “rakip” ama en iyi işbirliği/işveren adayı. Rakip tonu YOK — tamamlayıcı + hayran + meraklı.

4.3 Rheinwohnungsbau — Benjamin Gaidel (your most strategic contact)

Their angle: housing company, ~6,200 rental units (Düsseldorf/Duisburg/Meerbusch/Berlin); already energy-forward (Green Fusion heating, sector-coupling pilots). **Likely questions & answers:** - “What would you give us across 6,200 units?” → “A portfolio view: EUI and carbon per building, EPBD/MEPS risk, CRREM stranding year, and a ranked retrofit list with BAFA/KfW subsidy-adjusted ROI — i.e. which buildings to act on first.” - “Do you replace Techem/ista?” → “No, I sit above them and can use their export. I don’t do the billing; I do the portfolio decision and compliance layer.” - “How accurate is it?” → “Estimated with ranges now; it tightens with your real data, and I hand-check every figure before it reaches you.” **They’ll probe:** HKVO/UVI 2027, EPBD timelines, subsidy logic, effort required from them. **Discovery (don’t sell — ask):** “With EPBD coming, what’s hardest across your portfolio — knowing which buildings to act on first, the data, or the funding case?” **Now→future:** “A pilot would be me running a screening from your existing data; with funding I automate the connectors so it’s self-serve.” > TR — **Not:** Bu odadaki EN değerli sohbet. Satma, keşfet. Pilot = sana gerçek veri + referans; onlara düşük-eforlu EPBD/CapEx netliği.

4.4 BET Consulting — Leon Bücher

Their angle: large energy consultancy (100+ staff); his talk = smart-meter (iMSys) cost-benefit in multi-family + regulation to 2030. **Likely questions & answers:** - “How could this help our advisory?” → “Fast multi-building screening with transparent assumptions and branded, exportable reports — a tool that answers ‘which building, why’ quickly, even on incomplete data. White-label is on the table.” - “What methods/standards?” → “Archetype EUI, HDD normalization, ISO 13370 heat loss, GEG U-values, CRREM pathways, ERS-E1 — all transparent and auditable.” - “ML or rule-based?” → “Rule/physics-based and transparent by design — better for advisory and compliance defensibility.” **They’ll probe:** methodological rigor, standards, where your numbers come from. **Now→future:** “White-label/multi-tenant is partly built; I’d harden it for a

consulting partner.” > TR — **Not:** BET = pazar istihbaratı + olası white-label/freelance/iş. Onlar metodoloji sorar; Bölüm 2’yi iyi bil.

4.5 The Mobility House — Sven Böhme, Head of Trading

Their angle: V2G + flexibility trading on energy markets (“invisible power plant”); now pure-play flexibility (sold fleet/charging to Edenred). **Likely questions & answers:** - “Do you do flexibility/trading?” → “No — that’s your core. I’m the upstream building data and decision layer. The link is identifying flexibility potential early from a building’s existing data.” **They’ll probe:** whether you understand flexibility markets (aFRR/FCR, spot/intraday) — mostly listen and learn here. > TR — **Not:** İnce örtüşme, düşük öncelik. Öğren, iddia atma. “Monetizasyon/esneklik” temasını anlamak için iyi.

4.6 ZIA — Aygül Özkan (keynote, MD, former minister)

Their angle: Germany’s main real-estate association. Keynote: “regulatory pressure vs economic opportunity” = your exact thesis. **Approach:** nothing to sell; reference a concrete keynote point, connect on LinkedIn. Credibility + member network (housing companies). - *Question if you get a moment:* “You framed it as regulatory pressure vs opportunity — for smaller housing portfolios, which side actually makes them act first?” > TR — **Not:** Çok kıdemli, kalabalık. Hedef = kısa+saygılı + LinkedIn. Uzun sohbet bekleme.

5. Tough / awkward questions — honest answers

- “Do you have customers / real data?” → “Not yet — the platform is built and live, running on realistic synthetic data, and I’m lining up the first pilots. That’s exactly why a portfolio like yours is interesting to me.”
 - “You’re solo?” → “Yes, for now. I built the full platform myself — domain plus engineering. I’m open to the right partner who brings distribution.”
 - “Isn’t this just a dashboard?” → “The dashboard is the surface. The value is the estimation engine and the compliance/decision layer underneath — EPBD/CRREM risk and subsidy-adjusted prioritization.”
 - “Aren’t you competing with [metiundo / Green Fusion]?” → “No — they own the data / control layer; I’m the compliance and portfolio-decision layer above. Complementary.”
 - “Do customers need a Microsoft Fabric account?” → “No. They upload existing data; the heavy analytics run on my side. App-owns-data.”
 - “Why should I trust the numbers?” → “Every figure is a range with its assumptions, and I hand-check outputs on a new building before they’re used. It’s screening/decision-support, not stamped certification.” > TR — **Not:** Hiçbirinde savunmaya/özür moduna geçme. Dürüst + net + kısa. “Solo + sentetik veri” = zayıflık değil, dürüstçe söylenince olgunluk.
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6. Glossary — terms you’ll hear (own these)

Energy / buildings - EUI — Energy Use Intensity, kWh/m²/yr. (TR: enerji yoğunluğu) - **HDD / CDD** — Heating/Cooling Degree Days; weather normalization basis. (TR: ısıtma/soğutma dereceleri) - **U-value** — heat transfer through a building element, W/m²K (lower = better insulated). (TR: ısı geçirgenlik) - **COP / SCOP** — (Seasonal) Coefficient of Performance of a heat pump (heat

out ÷ electricity in). (TR: ısı pompası verim katsayısı) - **Sub-metering** — measuring consumption per unit/zone, not just the whole building. - **Load profile / peak shaving / load shifting** — shape of demand over time, and reducing/moving peaks.

German regulatory - GEG — Gebäudeenergiegesetz (Building Energy Act); **§71a** = building-automation duty. - **EPBD** — EU Energy Performance of Buildings Directive; source of **MEPS** (minimum standards). - **Energieausweis** — the EPC (energy certificate), classes A-H. - **HeizkostenV / HKVO + UVI** (unterjährliche Verbrauchsinformation) — monthly consumption info, mandatory from 2027. - **MsbG / iMSys / Smart Meter Gateway** — metering law / intelligent metering system / the secure gateway. - **Messstellenbetreiber** — metering point operator (what metiundo is). - **BAFA / KfW / BEG** — German subsidy bodies/programs for efficiency & heat pumps. - **Mieterstrom** — landlord-to-tenant on-site (PV) electricity model.

Carbon / ESG - Scope 1/2/3 — direct / purchased-energy / value-chain emissions. - **Location-based vs market-based** — grid-average vs contract/residual-mix Scope 2. - **Residual mix** — emission factor after green contracts are removed (higher). - **CRREM** — Carbon Risk Real Estate Monitor; decarbonization pathways & **stranding** year. - **ESRS-E1 / CSRD / VSME** — EU sustainability reporting standard / directive / voluntary SME standard. (Say “ESRS-E1-aligned,” not “CSRD-compliant.”) - **EU Taxonomy** — classification of “green” activities/assets. - **ETS2 / nEHS** — EU/national carbon pricing reaching buildings (~2028 / today). - **CO2KostAufG** — German law splitting the CO₂ cost between landlord and tenant.

Flexibility / market (this event’s theme) - Flexibility — ability to shift/shed load or feed in, sold on energy markets. - **Sektorkopplung** — sector coupling (heat + power + mobility together). - **V2G** — Vehicle-to-Grid; using EV batteries as grid storage. - **aFRR / FCR** — automatic frequency restoration / frequency containment reserve (balancing markets). - **Spot / intraday** — wholesale electricity markets (day-ahead / within-day). - **“Datengold” / Erlöspotenzial** — the event’s framing: data as the asset; revenue potential.

Tech / data - Medallion (bronze/silver/gold) — raw → cleaned → business-ready data layers. - **Microsoft Fabric / lakehouse / DirectLake** — the data platform / storage / fast query mode. - **RLS** — row-level security (each customer sees only their data). - **App-owns-data embed** — the customer doesn’t need their own BI/Fabric license.

TR — **Not:** Hepsini ezberleme; konuşmada GEÇENLERİ tanı + 4-5 tanesini rahat kullan (EUI, HDD, U-value, Scope 2 location/market, CRREM stranding, GEG §71a). Bu kadarı “ne yaptığını biliyor” izlenimi için yeter.